1 部分正确性注释程序

定义 1.1 (部分正确性注释程序)

对断言 P,Q, 程序 S 和某字符串 S^*, 递归定义

$$\{P\}\,S^*\,\{Q\}\ \vdash_{\mathrm{PO}}\ \{P\}\,S\,\{Q\},$$

称 $\{P\}\,S^*\,\{Q\}$ 是 $\{P\}\,S\,\{Q\}$ 的**注释程序**, $\{P\},\{Q\}$ 分别为前后断言 (pre, post), S 为骨架程序 (sk):

1. 当 $P\to Q$ 时, $\{P\}\,\mathrm{skip}\,\{Q\}\vdash_{\mathrm{PO}}\{P\}\,\mathrm{skip}\,\{Q\}$,
2. 如果有 Q_1 使得 $Q_1[u:=t]$ 是自由替换, $P\to Q_1[u:=t]$ 且 $Q_1\to Q$, 那么
$$\{P\}\,u:=t\,\{Q\}\vdash_{\mathrm{PO}}\{P\}\,u:=t\,\{Q\},$$
3. 如果 $\{P\}\,S_1^*\,\{R\}\vdash_{\mathrm{PO}}\{P\}\,S_1\,\{R\}$ 且 $\{R\}\,S_2^*\,\{Q\}\vdash_{\mathrm{PO}}\{R\}\,S_2\,\{Q\}$, 则
$$\{P\}\,S_1^*\,\{R\}\,S_2^*\,\{Q\}\vdash_{\mathrm{PO}}\{P\}\,S_1;S_2\,\{Q\},$$
4. 如果 $\{P\wedge B\}\,S_1^*\,\{Q\}\vdash_{\mathrm{PO}}\{P\wedge B\}\,S_1\,\{Q\}$ 且 $\{P\wedge\neg B\}\,S_2^*\,\{Q\}\vdash_{\mathrm{PO}}\{P\wedge\neg B\}\,S_2\,\{Q\}$, 则
$$\begin{aligned}&\{P\}\,\mathbf{if}\ B\ \mathbf{then}\ \{P\wedge B\}\,S_1^*\,\{Q\}\ \mathbf{else}\ \{P\wedge\neg B\}\,S_2^*\,\{Q\}\ \mathbf{fi}\ \{Q\}\\ \vdash_{\mathrm{PO}}\ &\{P\}\,\mathbf{if}\ B\ \mathbf{then}\ S_1\ \mathbf{else}\ S_2\ \mathbf{fi}\,\{Q\},\end{aligned}$$
5. 如果 $P\to I$, $\{I\wedge B\}\,S^*\,\{I\}\vdash_{\mathrm{PO}}\{I\wedge B\}\,S\,\{I\}$ 且 $I\wedge\neg B\to Q$, 则
$$\begin{aligned}&\{P\}\{\mathbf{inv}:I\}\,\mathbf{while}\ B\ \mathbf{do}\ \{I\wedge B\}\,S^*\,\{I\}\ \mathbf{od}\,\{Q\}\\ \vdash_{\mathrm{PO}}\ &\{P\}\,\mathbf{while}\ B\ \mathbf{do}\ S\ \mathbf{od}\,\{Q\}.\end{aligned}$$

定理 1.1 (正规形定理)

对任意的断言 P,Q 和程序 S, $\vdash_{\mathrm{PW}}\{P\}\,S\,\{Q\}$ 当且仅当存在 $\{P\}\,S^*\,\{Q\}\vdash_{\mathrm{PO}}\{P\}\,S\,\{Q\}$.

定义 1.2 (部分正确性注释程序上的操作)

把 $\{P\}\,\mathrm{E}\,\{P\}$ 也视为自身的注释程序, 递归定义注释程序带状态的一步转移关系:

1. $\langle\{P\}\,\mathrm{skip}\,\{Q\},\sigma\rangle\overset{\mathrm{def}}{\to}\langle\{Q\}\,\mathrm{E}\,\{Q\},\sigma\rangle$,
2. $\langle\{P\}\,u:=t\,\{Q\},\sigma\rangle\overset{\mathrm{def}}{\to}\langle\{Q\}\,\mathrm{E}\,\{Q\},\sigma[u:=\sigma(t)]\rangle$,
3. 任给注释程序 $\{P\}\,S_1^*\,\{R\}$ 和 $\{R\}\,S_2^*\,\{Q\}$, 如果 $\langle\{P\}\,S_1^*\,\{R\},\sigma\rangle\to\langle\{P'\}\,S_1'^{\,*}\,\{R\},\sigma'\rangle$, 那么
$$\langle\{P\}\,S_1^*\{R\}\,S_2^*\,\{Q\},\sigma\rangle\overset{\mathrm{def}}{\to}\langle\{P'\}\,S_1'^{\,*}\,\{R\}\,S_2^*\,\{Q\},\sigma'\rangle,$$
4. 任给注释程序 $\{P\}\,S_1^*\,\{R\}$ 和 $\{R\}\,S_2^*\,\{Q\}$, 如果 $\langle\{P\}\,S_1^*\,\{R\},\sigma\rangle\to\langle\{R\}\,\mathrm{E}\,\{R\},\sigma'\rangle$, 那么
$$\langle\{P\}\,S_1^*\{R\}\,S_2^*\,\{Q\},\sigma\rangle\overset{\mathrm{def}}{\to}\langle\{R\}\,S_2^*\,\{Q\},\sigma'\rangle,$$
5. 任给注释程序 $\{P\wedge B\}\,S_1^*\,\{Q\}$ 和 $\{P\wedge\neg B\}\,S_2^*\,\{Q\}$, 如果 $\sigma\models B$, 则
$$\langle\{P\}\,\mathbf{if}\ B\ \mathbf{then}\ \{P\wedge B\}\,S_1^*\,\{Q\}\ \mathbf{else}\ \{P\wedge\neg B\}\,S_2^*\,\{Q\}\ \mathbf{fi}\ \{Q\},\sigma\rangle\overset{\mathrm{def}}{\to}\langle\{P\wedge B\}\,S_1^*\,\{Q\},\sigma\rangle,$$
6. 任给注释程序 $\{P\wedge B\}\,S_1^*\,\{Q\}$ 和 $\{P\wedge\neg B\}\,S_2^*\,\{Q\}$, 如果 $\sigma\not\models B$, 则
$$\langle\{P\}\,\mathbf{if}\ B\ \mathbf{then}\ \{P\wedge B\}\,S_1^*\,\{Q\}\ \mathbf{else}\ \{P\wedge\neg B\}\,S_2^*\,\{Q\}\ \mathbf{fi}\ \{Q\},\sigma\rangle\overset{\mathrm{def}}{\to}\langle\{P\wedge\neg B\}\,S_2^*\,\{Q\},\sigma\rangle,$$
7. 任给 $P\to I$, 注释程序 $\{I\wedge B\}\,S^*\,\{I\}$ 和 $I\wedge\neg B\to Q$, 如果 $\sigma\models B$, 则
$$\begin{aligned}&\langle\{P\}\{\mathbf{inv}:I\}\,\mathbf{while}\ B\ \mathbf{do}\ \{I\wedge B\}\,S^*\,\{I\}\ \mathbf{od}\,\{Q\},\sigma\rangle\overset{\mathrm{def}}{\to}\\ &\langle\{I\wedge B\}\,S^*\,\{I\}\{\mathbf{inv}:I\}\,\mathbf{while}\ B\ \mathbf{do}\ \{I\wedge B\}\,S^*\,\{I\}\ \mathbf{od}\,\{Q\},\sigma\rangle,\end{aligned}$$
8. 任给 $P\to I$, 注释程序 $\{I\wedge B\}\,S^*\,\{I\}$ 和 $I\wedge\neg B\to Q$, 如果 $\sigma\not\models B$, 则
$$\langle\{P\}\{\mathbf{inv}:I\}\,\mathbf{while}\ B\ \mathbf{do}\ \{I\wedge B\}\,S^*\,\{I\}\ \mathbf{od}\,\{Q\},\sigma\rangle\overset{\mathrm{def}}{\to}\langle\{Q\}\,\mathrm{E}\,\{Q\},\sigma\rangle.$$

定理 1.2

对任意的 $\langle G,\sigma\rangle \to \langle G',\sigma'\rangle$,

1. $\mathrm{post}(G') = \mathrm{post}(G)$,
2. 如果 $\mathrm{sk}(G') = \mathrm{E}$, 那么 G 一定形如 $\{P\}\,\mathrm{E}\,\{P\}$,
3. $\langle \mathrm{sk}(G),\sigma\rangle \to \langle \mathrm{sk}(G'),\sigma'\rangle$,
4. 如果 $\sigma \models \mathrm{pre}(G)$, 那么 $\sigma' \models \mathrm{pre}(G')$.

证明.

1. 2. 显然. 3. 4. 对转移操作归纳证明.

i) $P \to Q$ 且 $G = \{P\}\,\mathrm{skip}\,\{Q\}$, 则 $G' = \{Q\}\,\mathrm{E}\,\{Q\}$, $\sigma' = \sigma$. 显然

$$\langle \mathrm{skip},\sigma\rangle \to \langle \mathrm{E},\sigma\rangle,$$

并且如果 $\sigma \models P$, 那么 $\sigma \models Q$.

ii) 有 Q_1 使得 $Q_1[u := t]$ 是自由替换, $P \to Q_1[u := t]$ 且 $Q_1 \to Q$, $G = \{P\}\,u := t\,\{Q\}$, 则

$$G' = \{Q\}\,\mathrm{E}\,\{Q\}, \quad \sigma' = \sigma[u := \sigma(t)],$$

显然 $\langle u := t,\sigma\rangle \to \langle \mathrm{E},\sigma[u := \sigma(t)]\rangle$, 并且如果 $\sigma \models P$, 那么 $\sigma \models Q_1[u := t]$, 从而 $\sigma' = \sigma[u := t] \models Q_1$, 即 $\sigma' \models Q$.

iii) $G = \{P\}\,S_1^*\,\{R\}\,S_2^*\{Q\}$, 假设 $\langle\{P\}\,S_1^*\,\{R\},\sigma\rangle \to \langle\{P'\}\,S_1'^{\,*}\,\{R\},\sigma'\rangle$ 并且

$$\langle S_1,\sigma\rangle \to \langle S_1',\sigma'\rangle, \quad \sigma \models P \to \sigma' \models P',$$

那么 $G' = \{P'\}\,S_1'^{\,*}\,\{R\}\,S_2^*\,\{Q\}$, 此时 $\langle S_1;S_2,\sigma\rangle \to \langle S_1';S_2,\sigma'\rangle$.

iv) $G = \{P\}\,S_1^*\,\{R\}\,S_2^*\{Q\}$, 假设 $\langle\{P\}\,S_1^*\,\{R\},\sigma\rangle \to \langle\{R\}\mathrm{E}\,\{R\},\sigma'\rangle$ 并且

$$\langle S_1,\sigma\rangle \to \langle \mathrm{E},\sigma'\rangle, \quad \sigma \models P \to \sigma' \models R,$$

那么 $G' = \{R\}\,S_2^*\,\{Q\}$, 此时 $\langle S_1;S_2,\sigma\rangle \to \langle S_2,\sigma'\rangle$.

v) $G = \{P\}\,\mathbf{if}\ B\ \mathbf{then}\ \{P\wedge B\}\,S_1^*\,\{Q\}\ \mathbf{else}\ \{P\wedge\neg B\}\,S_2^*\,\{Q\}\ \mathbf{fi}\ \{Q\}$ 且 $\sigma \models B$, 则

$$G' = \{P\wedge B\}\,S_1^*\,\{Q\}, \quad \sigma' = \sigma,$$

显然 $\langle\,\mathbf{if}\ B\ \mathbf{then}\ S_1\ \mathbf{else}\ S_2\ \mathbf{fi},\sigma\rangle \to \langle S_1,\sigma\rangle$, 并且如果 $\sigma \models P$, 那么 $\sigma' = \sigma \models P\wedge B$.

vi) $G = \{P\}\,\mathbf{if}\ B\ \mathbf{then}\ \{P\wedge B\}\,S_1^*\,\{Q\}\ \mathbf{else}\ \{P\wedge\neg B\}\,S_2^*\,\{Q\}\ \mathbf{fi}\ \{Q\}$ 且 $\sigma \not\models B$, 则

$$G' = \{P\wedge\neg B\}\,S_2^*\,\{Q\}, \quad \sigma' = \sigma,$$

显然 $\langle\,\mathbf{if}\ B\ \mathbf{then}\ S_1\ \mathbf{else}\ S_2\ \mathbf{fi},\sigma\rangle \to \langle S_2,\sigma\rangle$, 并且如果 $\sigma \models P$, 那么 $\sigma' = \sigma \models P\wedge\neg B$.

vii) $P \to I$, $I\wedge\neg B \to Q$, $G = \{P\}\{\mathbf{inv}:I\}\,\mathbf{while}\ B\ \mathbf{do}\ \{I\wedge B\}\,S^*\,\{I\}\ \mathbf{od}\,\{Q\}$ 且 $\sigma \models B$, 则

$$G' = \{I\wedge B\}\,S^*\,\{I\}\{\mathbf{inv}:I\}\,\mathbf{while}\ B\ \mathbf{do}\ \{I\wedge B\}\,S^*\,\{I\}\ \mathbf{od}\,\{Q\}, \quad \sigma' = \sigma,$$

显然

$$\langle\,\mathbf{while}\ B\ \mathbf{do}\ S\ \mathbf{od},\sigma\rangle \to \langle S;\mathbf{while}\ B\ \mathbf{do}\ S\ \mathbf{od},\sigma\rangle,$$

并且如果 $\sigma \models P$, 那么 $\sigma \models I$, 从而 $\sigma' = \sigma \models I\wedge B$.

viii) $P \to I$, $I\wedge\neg B \to Q$, $G = \{P\}\{\mathbf{inv}:I\}\,\mathbf{while}\ B\ \mathbf{do}\ \{I\wedge B\}\,S^*\,\{I\}\ \mathbf{od}\,\{Q\}$ 且 $\sigma \not\models B$, 则

$$G' = \{Q\}\,\mathrm{E}\,\{Q\}, \quad \sigma' = \sigma,$$

显然

$$\langle\,\mathbf{while}\ B\ \mathbf{do}\ S\ \mathbf{od},\sigma\rangle \to \langle \mathrm{E},\sigma\rangle,$$

并且如果 $\sigma \models P$, 那么 $\sigma \models I$, 从而 $\sigma' = \sigma \models I\wedge\neg B$, 即 $\sigma' \models Q$. □

2 完全正确性注释程序

定义 2.1 (完全正确性注释程序)

对断言 P, Q, 程序 S 和某字符串 S^*, 递归定义

$$\{P\}\, S^*\, \{Q\} \ \vdash_{\mathrm{TO}} \ [P]\, S\, [Q],$$

称 $\{P\}\, S^*\, \{Q\}$ 是 $[P]\, S\, [Q]$ 的**完全正确性注释程序**, 前后断言与骨架程序的记法 $\mathrm{pre}, \mathrm{post}, \mathrm{sk}$ 同上节:

1. 当 $P \to Q$ 时, $\{P\}\, \mathrm{skip}\, \{Q\} \vdash_{\mathrm{TO}} [P]\, \mathrm{skip}\, [Q]$,
2. 如果有 Q_1 使得 $Q_1[u := t]$ 是自由替换, $P \to Q_1[u := t]$ 且 $Q_1 \to Q$, 那么
$$\{P\}\, u := t\, \{Q\} \vdash_{\mathrm{TO}} [P]\, u := t\, [Q],$$
3. 如果 $\{P\}\, S_1^*\, \{R\} \vdash_{\mathrm{TO}} [P]\, S_1\, [R]$ 且 $\{R\}\, S_2^*\, \{Q\} \vdash_{\mathrm{TO}} [R]\, S_2\, [Q]$, 则
$$\{P\}\, S_1^*\, \{R\}\, S_2^*\, \{Q\} \vdash_{\mathrm{TO}} [P]\, S_1; S_2\, [Q],$$
4. 如果 $\{P \wedge B\}\, S_1^*\, \{Q\} \vdash_{\mathrm{TO}} [P \wedge B]\, S_1\, [Q]$ 且 $\{P \wedge \neg B\}\, S_2^*\, \{Q\} \vdash_{\mathrm{TO}} [P \wedge \neg B]\, S_2\, [Q]$, 则
$$\begin{aligned}&\{P\}\, \mathbf{if}\ B\ \mathbf{then}\ \{P \wedge B\}\, S_1^*\, \{Q\}\ \mathbf{else}\ \{P \wedge \neg B\}\, S_2^*\, \{Q\}\ \mathbf{fi}\ \{Q\}\\ &\vdash_{\mathrm{TO}} [P]\, \mathbf{if}\ B\ \mathbf{then}\ S_1\ \mathbf{else}\ S_2\ \mathbf{fi}\, [Q],\end{aligned}$$
5. 如果 $P \to I$, $\{I \wedge B\}\, S^*\, \{I\} \vdash_{\mathrm{TO}} [I \wedge B]\, S\, [I]$, t, z 是整型变量, z 不出现在 t, B, I, S 中, $\{I \wedge B \wedge t = z\}\, S^{**}\, \{t < z\} \vdash_{\mathrm{TO}} [I \wedge B \wedge t = z]\, S\, [t < z]$, $I \to t \geqslant 0$ 且 $I \wedge \neg B \to Q$, 则
$$\begin{aligned}&\{P\}\{\mathbf{inv} : I\}\{\mathbf{bd} : t\}\, \mathbf{while}\ B\ \mathbf{do}\ \{I \wedge B\}\, S^*\, \{I\}\ \mathbf{od}\, \{Q\}\\ &\vdash_{\mathrm{TO}} \ [P]\, \mathbf{while}\ B\ \mathbf{do}\ S\ \mathbf{od}\, [Q].\end{aligned}$$

定理 2.1 (正规形定理)

对任意的断言 P, Q 和程序 S, $\vdash_{\mathrm{TW}} [P]\, S\, [Q]$ 当且仅当存在 $\{P\}\, S^*\, \{Q\} \vdash_{\mathrm{TO}} [P]\, S\, [Q]$.

完全正确性注释程序上的操作几乎与定义 1.1 相同 (只是多了 **bd** 标记), 定理 1.2 的四条性质对完全正确性注释程序也依旧成立.